GLOBAL ENERGY HACKATHON 2026 · KESHO AI

Kesho AI: pitch video script (PowerPoint, 5 slides, 5:00)

Present Kesho_AI_Pitch.pptx in slide-show mode and screen-record with your voice (camera on if you can). About 130 words a minute. Times are a guide.

Slide 1 · Title (0:00 to 0:40)

"Hi, we're Kesho AI: Fereshteh Jafari, Sai Karri and Nworah Austin. [Each person: one line on your role.] We chose Brief 03: predict mini-grid demand before building, in the Digital Solutions and AI track. Kesho means 'tomorrow' in Swahili, because we show a village's power needs before anyone builds there. Kesho AI forecasts demand from satellite roofs and open data, then right-sizes and finances the solar mini-grid, for developers, lenders and electrification agencies."

Slide 2 · Impact (0:40 to 1:50)

"About 490 million people need mini-grids by 2030. That's 217,000 grids and about 127 billion dollars, but at today's pace only about 44,800 get built. Every one is sized from guesses: old census data, or surveys asking people how much power they'll use when they've never had it. Measured use runs from 5 to 35 kWh per customer a month, and forecasts are often overstated. Build too big and the tariff prices people out. Too small and the lights go out every evening.

Across our six villages in six countries, 2,888 households connect by year five, each with a forecast range, not a single guess. A single build for the middle forecast has a 37% chance of brownouts by year five; our staged plan holds that to 15%, with far less idle equipment. And at Gbamu Gbamu in Nigeria, sharing one plant with nearby settlements saves about 1.1 million dollars."

Slide 3 · Innovation (1:50 to 2:55)

"Today, one system is built for year five, so it sits idle for the first years and browns out later. The root cause: there is no demand data before building, so the risk lands on the village and the lender.

Our approach works in three steps. One: count and classify every roof from Google, Microsoft and OpenStreetMap footprints, checked against fresh Sentinel-2 satellite pictures. Two: forecast five years of demand across 4,000 possible futures, pulled toward real meter data from mini-grids within 1,000 miles, and tested on ten years of real weather and climate outlooks. Three: build in right-sized stages, price the wires or a grid line, and find the grant or loan that makes it pay back."

Slide 4 · Sustainability and affordability (2:55 to 4:05)

"For a household, power costs about 5 to 8 dollars a month at 85 cents a kWh. That's inside the 5 to 20 dollars families already spend on kerosene, candles and batteries.

Who pays? Households pay for what they use. Results-based grants, about 433 dollars per connection, and concessional loans cover the build, and Kesho AI finds the smallest grant needed. Bills cover running costs and the loan from year two, and each stage is added only when the meters show it's needed. With staging, today's equipment prices and a development-bank loan, the owner gets their money back in about four to six years, with no ongoing grants. Local technicians run it, battery renewal around year 11 is in the 20-year plan, and at Gbamu Gbamu alone the system makes about 290 megawatt-hours of solar a year, replacing kerosene and diesel."

Slide 5 · Feasibility and scale (4:05 to 5:00)

"This is a working prototype on six real villages, built this weekend, on open data only. The best evidence: at Gbamu Gbamu a real 85 kWp plant was built, and it matches our year-two demand of 84 kWp. We tested our

building classes on 2,225 tagged buildings. Our biggest risk is that these villages have no meters yet, and a one-day household count fixes most of that.

Next: a pilot at Gbamu Gbamu, building on the existing plant. Then nearby villages and sites in Nigeria's REA DARES and SEforALL's Universal Energy Facility grant programmes. Then any village in Africa, because every dataset covers the continent.

Our ask: a pilot developer, meter data from 50 to 200 sites, and an introduction to REA Nigeria. Thank you."